Q4		
(a)	From Fig. 4.2, 5 mm corresponds to 3 fringe separations. $3\Delta y = 5.0 \Rightarrow \Delta y = 5.0 \div 3 = 1.67$ mm $\Delta y = \dfrac{L\lambda}{d}$ $1.67\times10^{-3} = \dfrac{0.980(633\times10^{-9})}{d}$ $d = 3.7\times10^{-4}$ m	
(b)	Let θ be the angle made by 1st order minimum of the single slit diffraction envelope from the principle axis. From Fig. 4.1 and Fig 4.2, $\tan\theta_{\text{min}} = \dfrac{10.0\times10^{-3}}{0.980} = 0.0102$ $\sin\theta_{\text{min}} = \dfrac{\lambda}{b}$ $b = \dfrac{\lambda}{\sin\theta} \simeq \dfrac{\lambda}{\theta}$ (Applying small angle approximation) $= \dfrac{633\times10^{-9}}{0.0102}$ $= 6.2\times10^{-5}$ m <u>Alternative Method</u> From Fig. 4.2, the first minimum for single slit diffraction envelope coincides with the 6th maximum for double slit interference (missing order). Single slit diffraction: $b\sin\theta_1 = 1\lambda$ Double slit interference: $d\sin\theta_6 = 6\lambda$ Since $\theta_1 = \theta_6$, $\dfrac{b}{d} = \dfrac{1}{6}$ $b = \dfrac{d}{6} = \dfrac{3.7\times10^{-4}}{6}$ $= 6.2\times10^{-5}$ m	
(c)(i)	Increase slit separation *d*. From $\Delta y = \dfrac{\lambda L}{d}$, a smaller *d* makes the fringe separation Δy wider, spreading the interference pattern.	
(c)(ii)	Make the slits narrower. Narrower slits produce a much wider single-slit diffraction envelope with lower overall intensity, so the bright fringes vary less in intensity.	